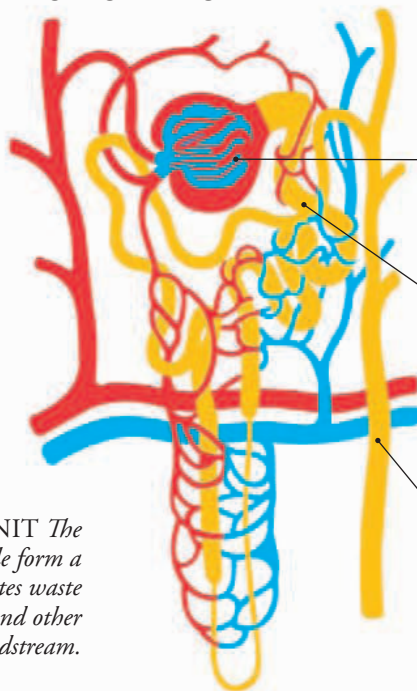


The capsule forms a protective coat around the kidney.

Each urine-making nephron consists of a cluster of blood capillaries surrounded by a capsule that leads to a long, winding tubule (tiny tube). Fluid, without blood cells, passes from blood flowing through the bundle of capillaries into the capsule and tubule. As the filtered fluid travels down and up the tubule, most of the water and other useful substances that the body needs, such as glucose (a sugar that gives us energy), are taken back into the bloodstream. What remains—excess water and salt, and waste substances—forms urine.

► **FILTRATION UNIT** *The capillaries and the tubule form a filtration unit that separates waste matter and returns water and other essential substances to the bloodstream.*

HOW URINE IS MADE



As blood flows through a cluster of capillaries, fluid passes out of it into a tubule (yellow).

The tubule twists and turns as it travels downward and then upward to join the collecting duct.

The collecting duct (tube) drains urine from many nephrons into the hollow pelvis of the kidney.

WHAT'S IN URINE?

Your urine is made up of water and dissolved substances that the body needs to get rid of. Urine usually has a yellow tint due to a chemical from worn-out blood cells, which the kidneys filter out of the bloodstream.

It is the same chemical that makes bruises yellow.



Water makes up 94 percent of urine.

Urea is produced in the liver when proteins are broken down into amino acids.

Sodium (excess salt) is also disposed of in urine.

Urine also contains small amounts of other dissolved substances.



TESTING URINE

A doctor may test urine to help find out what is wrong with a patient. A dipstick with several colored bands is dipped in a sample of the patient's urine. The bands change color when they react with certain chemicals in urine. These are then matched against a color chart to detect abnormal levels of any substance in the urine sample.

